

Chaitanya Kothapalli

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EDUCATION

- Santa Clara University** Santa Clara, CA
• *Masters in Computer Science and Engineering, CGPA: 3.6* Jan 2023 - Dec 2024
Courses: Design and Analysis: Algorithms, Computer Networks, Operating Systems, CA, Database Systems, Distributed Systems
- B V Raju Institute of Technology** India
• *Bachelors in Computer Science and Engineering, CGPA: 3.5* Jun 2017 - Jul 2021
Courses: Object Oriented Analysis and Design, Socket Programming, Cloud Computing, Compiler Design, Machine Learning, Web Tech

PROFESSIONAL EXPERIENCE

- Frugal Innovation Hub - Santa Clara University** Research Work
• *Java Backend Developer* Jan 2025 - Present
 - Developing backend services for a Library Management System using Spring Boot, Spring Data JPA, and REST APIs to manage users, book lending, and inventory with role-based access control and MySQL integration.
- Tata Consultancy Services** Full-Time Employment
• *Software Engineer-Backend* Aug 2021 - Sep 2022
Technologies: Java, Spring Boot, Restful API's, Microservices, Docker, Amazon EKS
 - Developed and deployed Java based microservices using Spring Boot for managing Active Directory groups in a Docker containerized environment Amazon Elastic Kubernetes Service , achieving a successful implementation with a 95 percent reduction in manual group management efforts.
 - Reduced cost by 10 percent by optimizing code for 22 active directory groups issues under External Account Maintenance and Bill Pay having 4 lakh users for United Services Automobile Association.
 - Delivered a comprehensive full-stack application following software development life cycle with Java, Python, React JS, and highly responsive RESTful APIs, incorporating rigorous data quality checks and advanced data governance for managing over 215 internal big data apps.
 - Engineered high utility microservices based application, catering to over 3500 stakeholders, by utilizing Spring Boot and MariaDB and ensuring seamless continuous deployment by implementing reliable DevOps Pipelines.
 - Simplified query optimization and in-memory Redis caching speeding up data processing by 33 percent for stakeholders' ML models and analytics dashboard.
- Larsen and Toubro Technology Services** Internship
• *Software Engineer-Data Intern* June 2021 - Aug 2021
Technologies: Java, Kafka, Scala, SQL, Spark.
 - Employed ETL pipelines in PySpark and Scala, running within a Kubernetes cluster harmonizing data from financial institutions, regulators, market data providers, and external big data sources, elevating data quality and delivery by 69 percent.
 - Improved query performance and transaction time by 2.5x through SQL caching and indexing techniques.

PROJECTS

- Bronco Career Alerts - Java, Python, RabbitMQ, Docker, Kubernetes, Distributed Systems:** It's a distributed system Pub-Sub model that lets students to get updates regarding job openings. Implemented this using algorithms like Multi casting using Message Passing Interface RabbitMQ, Replication using Docker and Kubernetes, Mutual Exclusion and Concurrency Control for Synchronization and deadlock prevention by using python, flask,integrated with External API's.
- House Leasing Application - Spring Boot, MongoDB, AWS S3, Docker, MVC, Security, Restful API's, OOAD:** Developed a comprehensive house leasing application with user and admin authentication, dynamic house listings featuring detailed maps and images, and a lease management system with approval workflows. Utilized Spring Boot for backend development, MongoDB for data management, and AWS S3 for secure file storage. Employed Docker for containerized deployment and MVC design pattern for robust architecture.
- Recursive DNS Resolver - Golang, DNS, Caching, Web Architecture and Protocols :** Developed a recursive DNS resolver in Golang that handles A, AAAA, NS, CNAME, and TXT record types. It resolves IPv4 and IPv6 addresses, name servers, and canonical names, including the A record for CNAMEs. Features a caching mechanism using TTL to reduce latency and network overhead, and supports TXT records retrieval by generating random numbers within a specified range.
- CI/CD EKS Fargate - ALB, VPC, IAM, Code Pipeline, Code Build, AWS EKS:** Designed and implemented a scalable cloud infrastructure on AWS using CodeCommit, CodeBuild, CodePipeline, and ECR for CI/CD, orchestrated containerized workloads with EKS Fargate Serverless Compute, and managed network traffic with a VPC and ALB.

SKILLS SUMMARY

- Programming Languages:** Java, Golang, C/C++, Python, JavaScript, SQL, Linux(OS)
- Frameworks:** Spring Boot (Spring MVC, Spring Data JPA, Hibernate), Kafka, RabbitMQ, Flask, Spark
- Databases:** MySQL, MongoDB, PostgreSQL, Cassandra, MariaDB, Redis
- Tools:** JIRA, PostMan, GIT, Jenkins, PuTTY, Docker, MiniKube, Locust, Jmeter

CERTIFICATIONS AND CO-CURRICULAR ACTIVITIES

- Certified in AWS Cloud Practitioner.
- Serving as a Gameday marketing member for Santa Clara University Athletics Department.
- Hosted a Technical Fest(Promethean) as Lead Organizer at BVRIT college.